How Digital Workers See & Control Computers

Accessibility-Tree Snapshots: Semantic Understanding Over Brittle Visual Pixels

The Semantic Revolution in Browser Automation

Traditional browser automation fails because web pages frequently change CSS classes, responsive layouts, and visual styling. Fathom operates browsers via the **Accessibility Tree** (the standard semantic model used by screen readers).

LEGACY PIXEL / CSS AUTOMATION (FRAGILE)

- **Obfuscated CSS:** Class names change on every frontend software build.
- **Pixel Drift:** Screen scaling causes mouse clicks to hit empty whitespace.
- **High Latency:** Uploading screenshots consumes massive bandwidth and tokens.

FATHOM SEMANTIC ACCESSIBILITY (ROCK SOLID)

- **Semantic Roles:** The agent sees functional elements: `Button: "Login"`.
- **Opaque Numerical Refs:** Direct addressing via stable tokens (e.g. `@e14`).
- **10x Token Efficiency:** Lightweight snapshots use 90% fewer tokens.

Accessibility Snapshot Architecture

1. Chromium Page	→	2. ARIA Snapshot	→	3. Opaque Tokens	→	4. Agent Action
Dynamic DOM & SPAs		Extracts semantic tree		Assigns stable refs (@e1)		click(@e1), type(@e2)

Layout-Agnostic Stability: Whether a website is redesigned, translated into Japanese, or resized to mobile viewport, the accessibility tree retains functional semantic integrity.